# diffpair_test_routed

Design Report

Rev 1 | 2026-11-12 | jlcpcb-tier1

kicad-tools 0.29.0

## Board Summary

| Property | Value |
|---|---|
| Layers | 4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu) |
| Footprints | 39 (36 SMD, 3 THT, 0 other) |
| Nets | 18 |
| Traces | 183 segments |
| Vias | 55 |
| Board Size | 90.0 x 70.0 mm |

# Stackup

| Layer | Type | Thickness (mm) | Material |
|---|---|---|---|
| F.Cu | copper | 0.035 | -- |
| dielectric 1 | prepreg | 0.2104 | 7628 |
| In1.Cu | copper | 0.0152 | -- |
| dielectric 2 | core | 1.065 | FR4 |
| In2.Cu | copper | 0.0152 | -- |
| dielectric 3 | prepreg | 0.2104 | 7628 |
| B.Cu | copper | 0.035 | -- |

# Design Overview

## Theory of Operation

Four-channel LVDS link demonstrator

# ERC Status

| Metric | Count |
|---|---|
| Errors | 0 |
| Warnings | 0 |

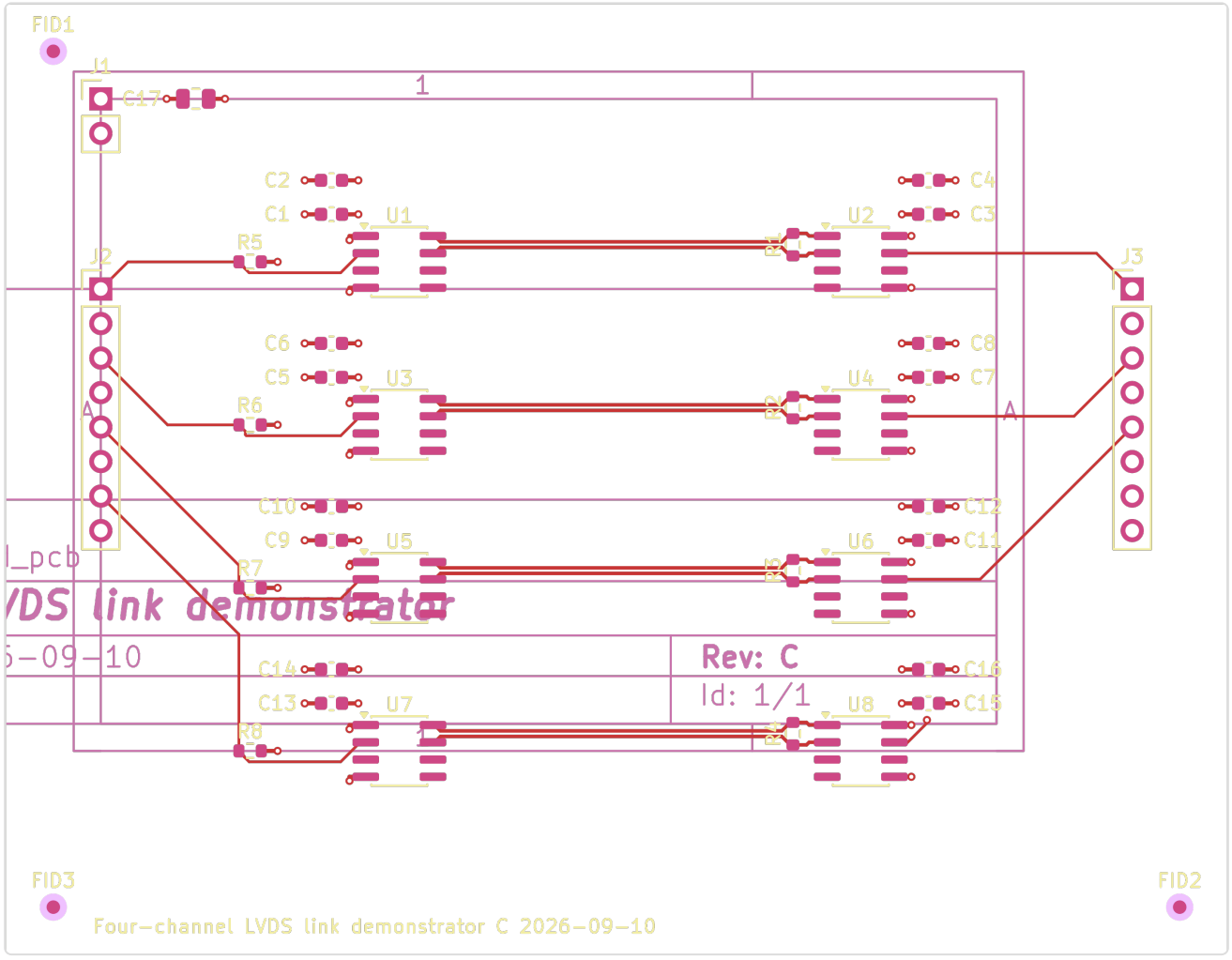
**Status**: PASS

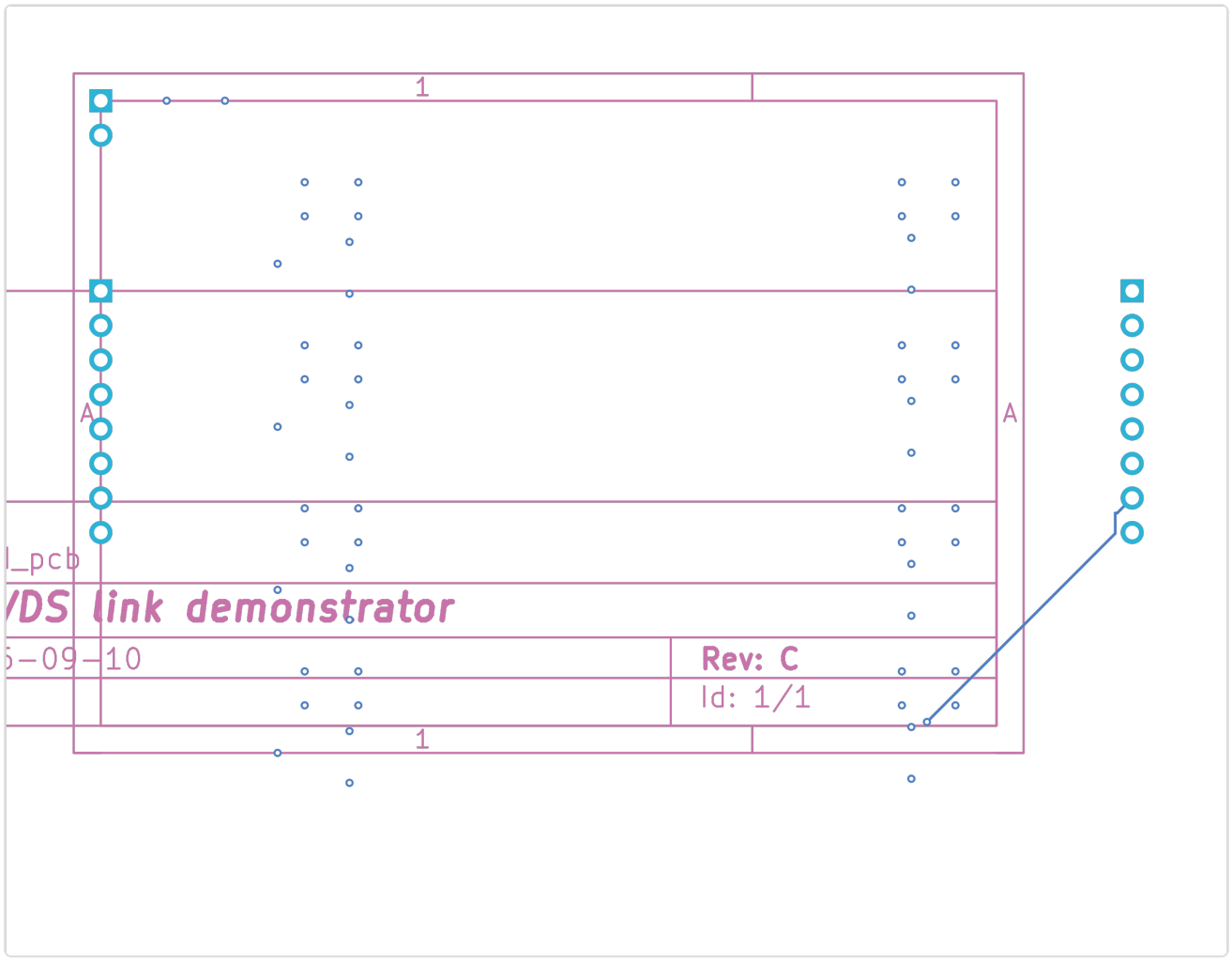
# Schematic Overview

## Schematic: diffpair_test

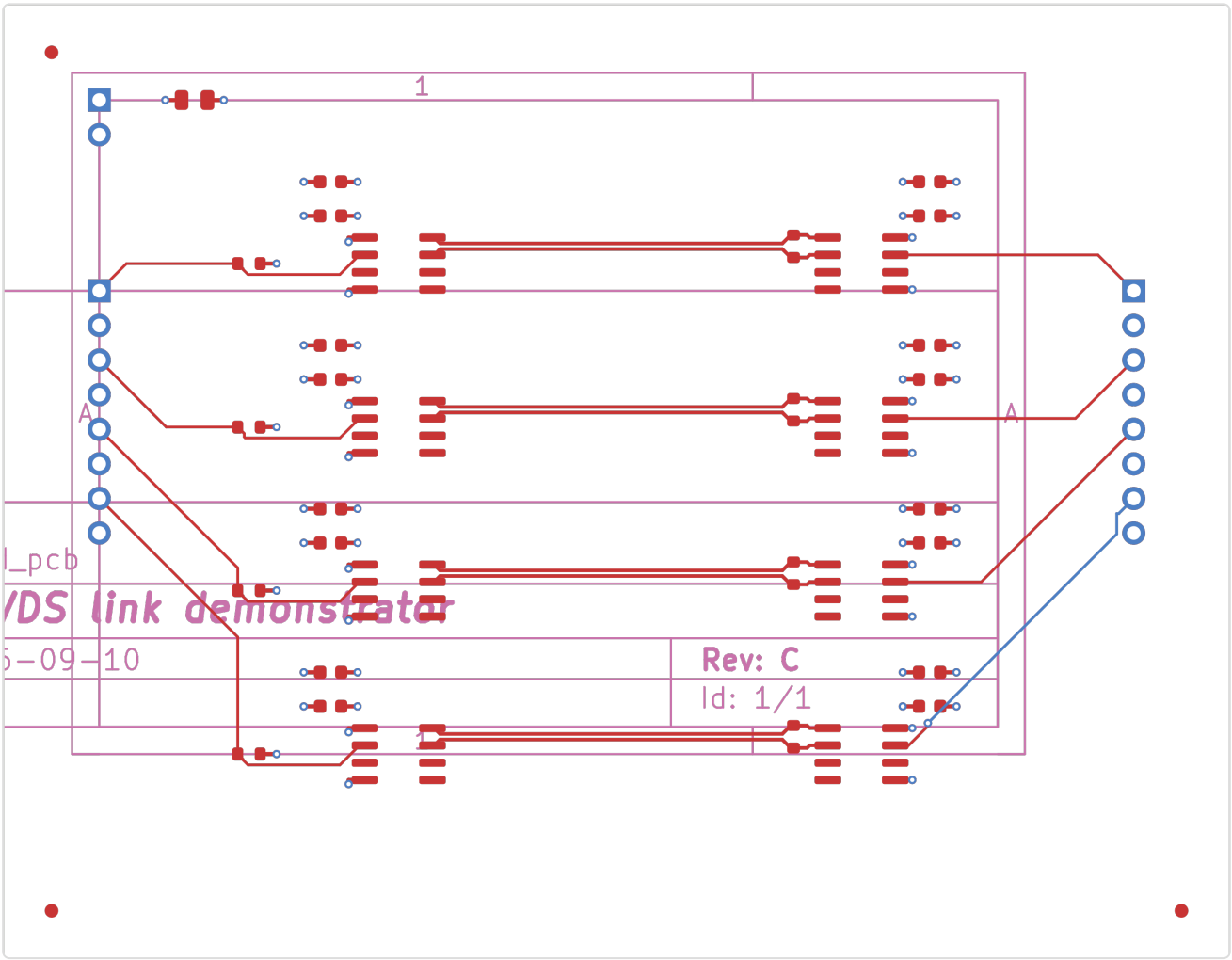


# PCB Layout

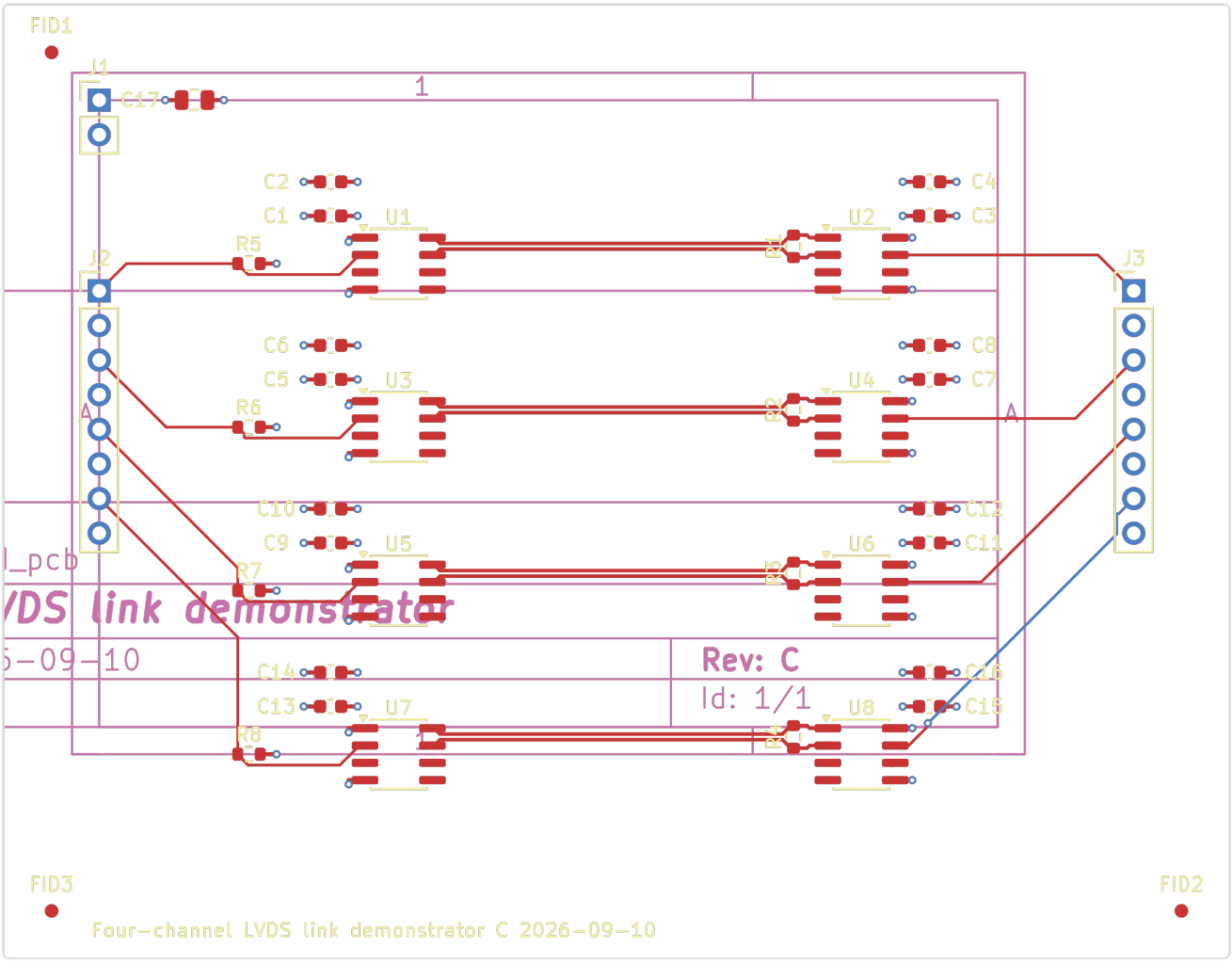

FID1
J1
C17
1
C2
C1
U1
R5
J2
C4
C3
U2
J3
C6
C5
U3
R6
A
C8
C7
U4
A
C10
C9
U5
R7
C12
C11
U6
l_pcb
/DS link demonstrator
6-09-10
Rev: C
Id: 1/1
C14
C13
U7
R8
C16
C15
U8
FID3
FID2
Four-channel LVDS link demonstrator C 2026-09-10


1
A
A
l_pcb
/DS link demonstrator
6-09-10
Rev: C
Id: 1/1
1

## Copper

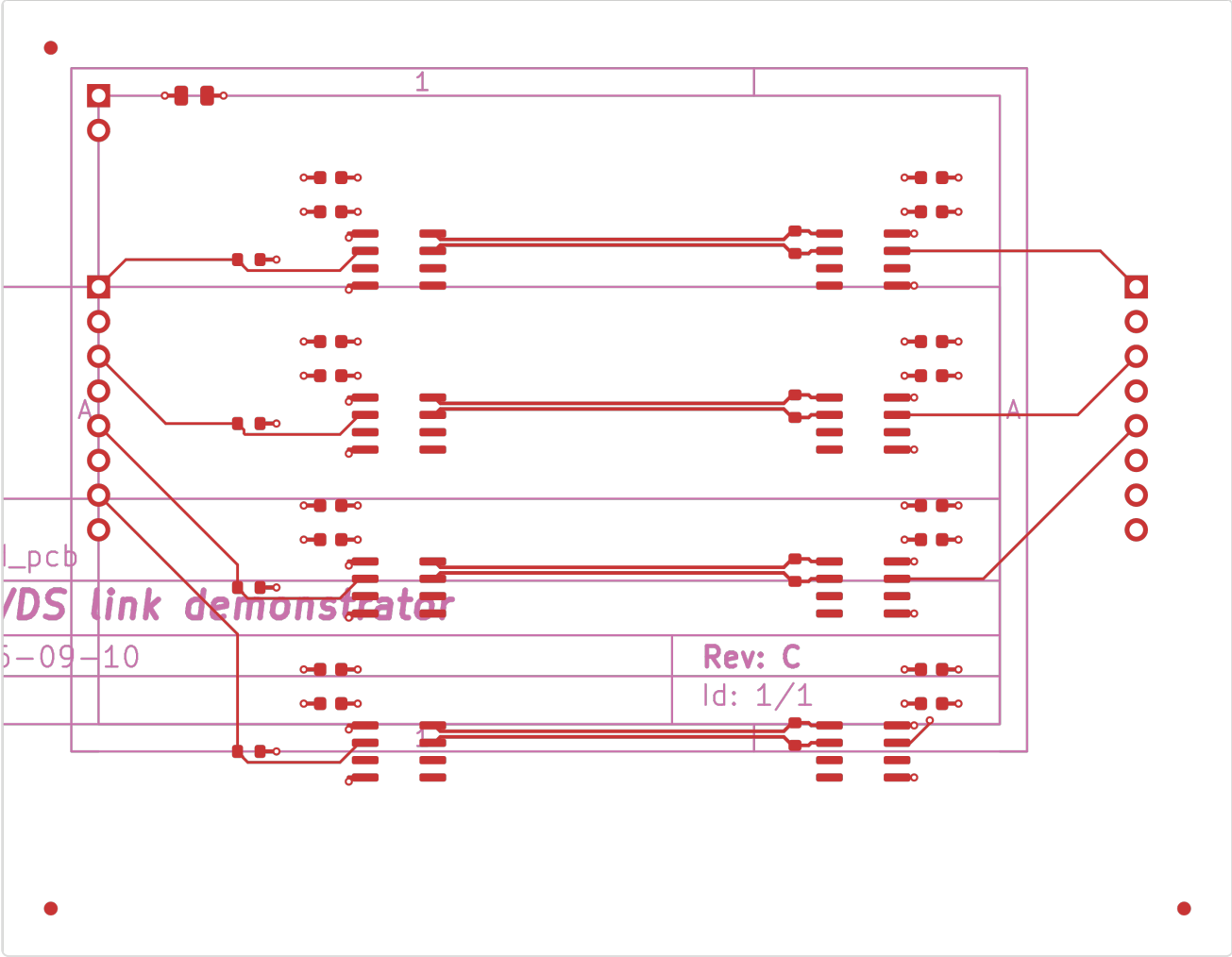


## Assembly

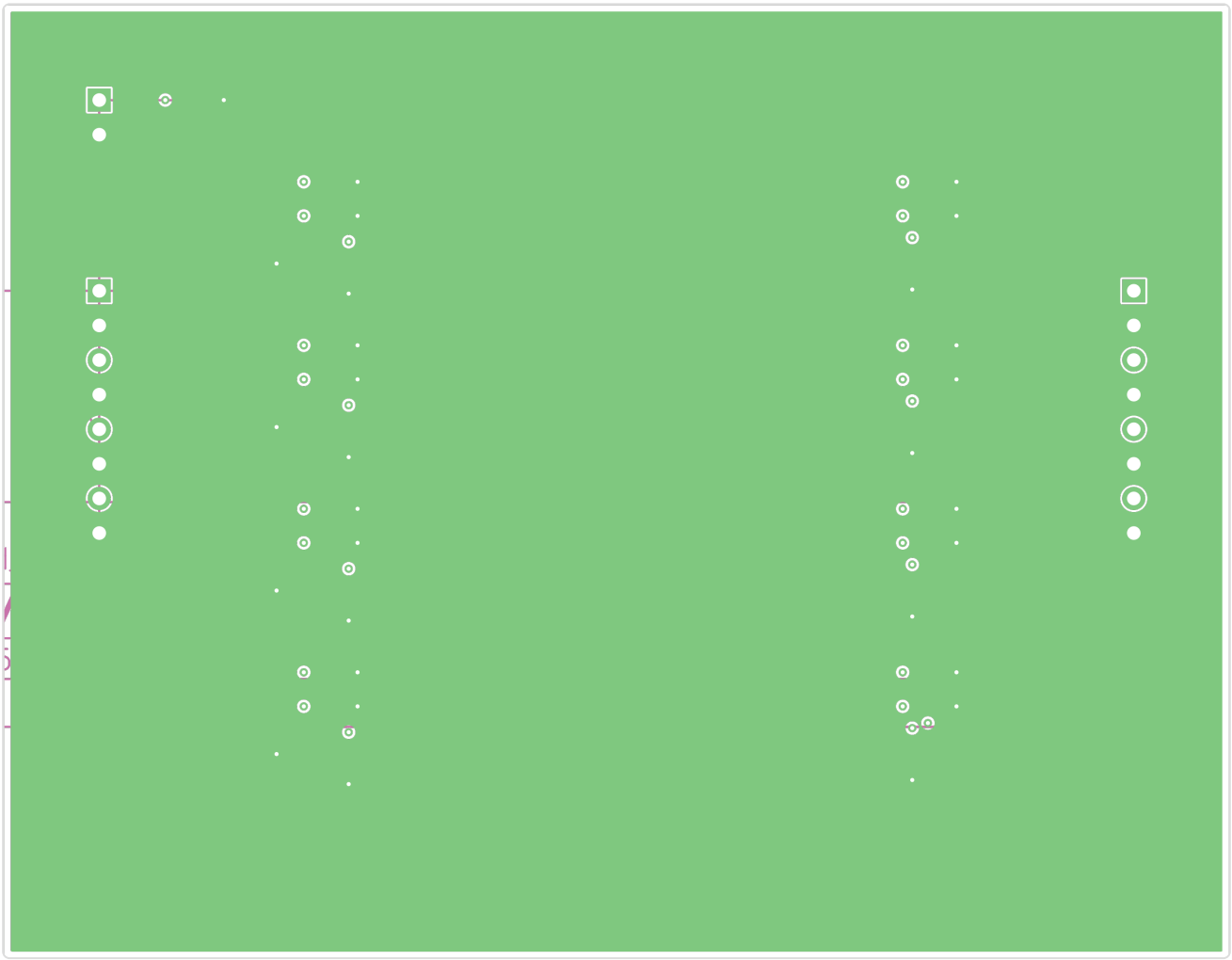

FID1
J1
C17
1
C2
C1
U1
R5
J2
C4
C3
U2
J3
C6
C5
U3
R6
A
C8
C7
U4
A
C10
C9
U5
R7
C12
C11
U6
l_pcb
/DS link demonstrator
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Rev: C
Id: 1/1
C14
C13
U7
R8
C16
C15
U8
FID3
FID2
Four-channel LVDS link demonstrator C 2026-09-10

# Copper Layers

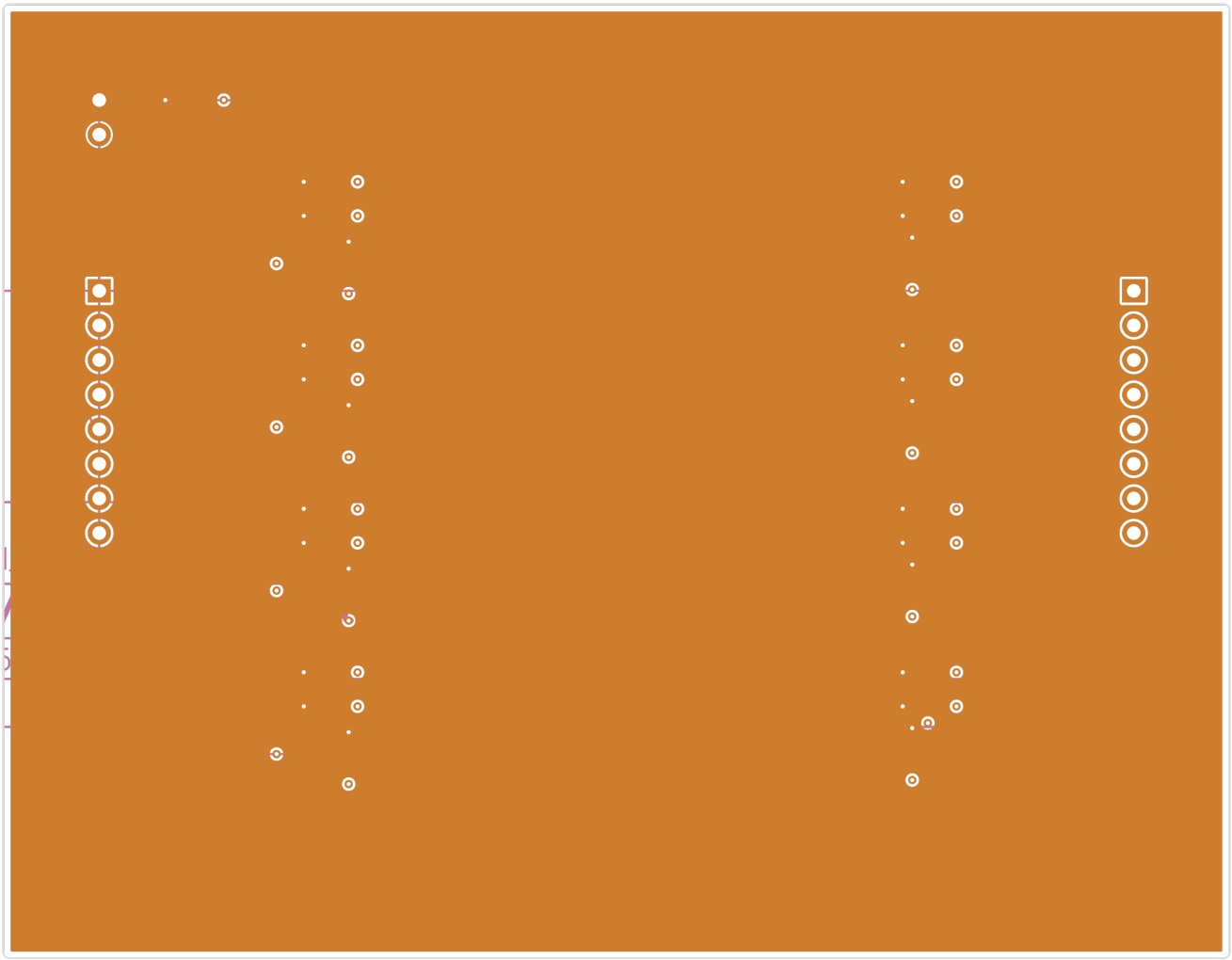
## F.Cu



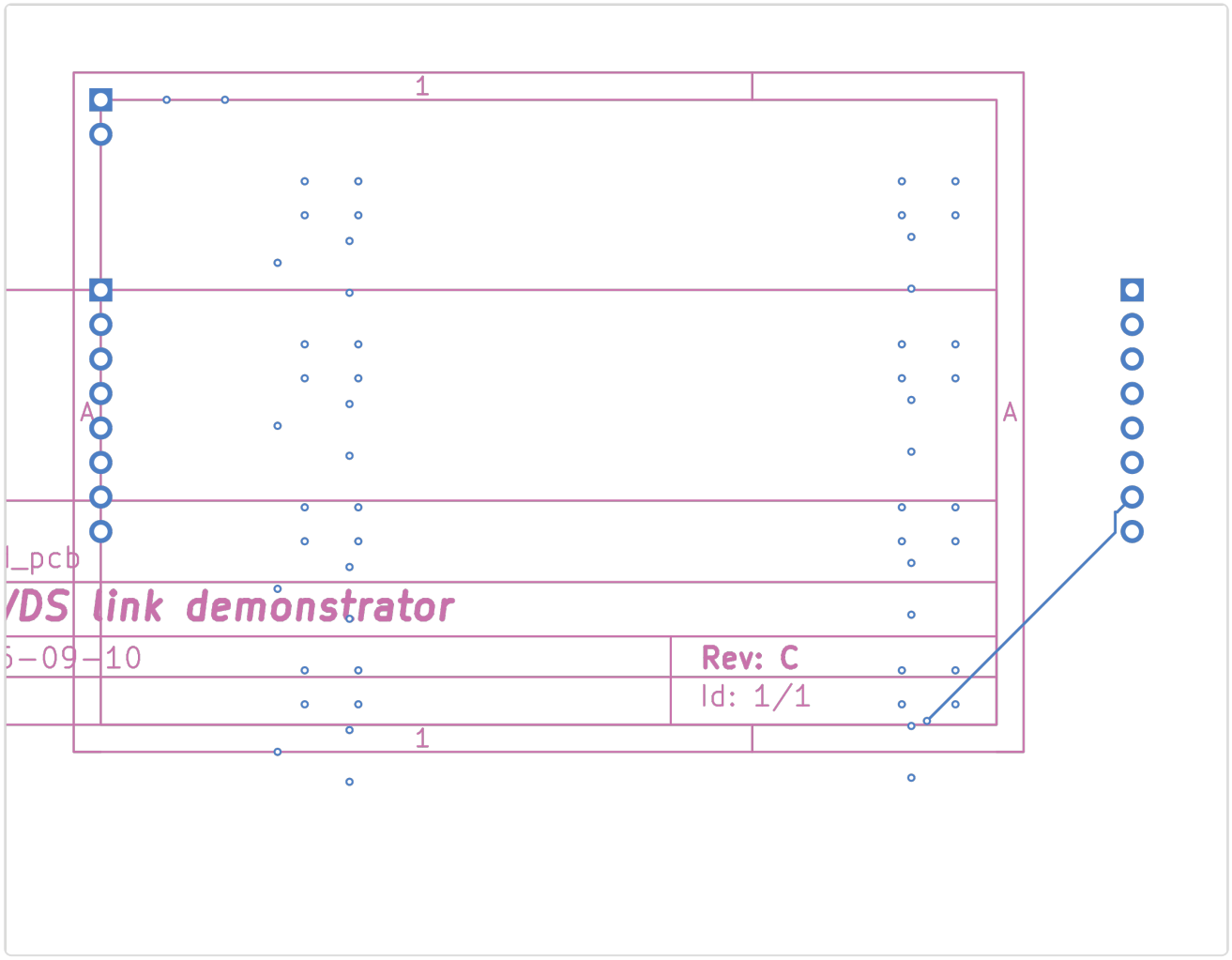
## In1.Cu



## In2.Cu



## B.Cu



# Bill of Materials

| Value | Package | Qty | References |
|---|---|---|---|
| 100nF | C_0603_1608Metric | 8 | C2, C4, C6, C8, C10, C12, C14, C16 |
| 10uF | C_0805_2012Metric | 1 | C17 |
| 1nF | C_0603_1608Metric | 8 | C1, C3, C5, C7, C9, C11, C13, C15 |
| 3.3V regulated input | PinHeader_1x02_P2.54mm_Vertical | 1 | J1 |
| IN1-4 / GND | PinHeader_1x08_P2.54mm_Vertical | 1 | J2 |
| OUT1-4 / GND | PinHeader_1x08_P2.54mm_Vertical | 1 | J3 |
| 100 | R_0603_1608Metric | 4 | R1, R2, R3, R4 |
| 100k | R_0603_1608Metric | 4 | R5, R6, R7, R8 |
| SN65LVDS1DR | SOIC-8_3.9x4.9mm_P1.27mm | 4 | U1, U3, U5, U7 |
| SN65LVDS2DR | SOIC-8_3.9x4.9mm_P1.27mm | 4 | U2, U4, U6, U8 |

# DRC Status (`--mfr jlcpcb-tier1` rules)

| Metric | Count |
|---|---|
| Errors | 0 |
| Errors: kct (`--mfr jlcpcb-tier1` rules) | 0 |
| Errors: kicad-cli (geometric) | 0 |
| Warnings | 0 |
| Blocking | 0 |

**Status**: PASS

# Manufacturing Readiness

**Verdict**: READY

## Action Items

- **[OPTIONAL]** Refill zones to verify 2 zone-connected nets, then re-run kct audit

# Routing Status

| Metric | Value |
|---|---|
| Signal Net Completion | 100.0% (16/16) |
| Overall Completion | 100.0% |
| Complete Nets | 18 / 18 |
| Zone-Connected Nets | 2 |
| Incomplete Nets | 0 |
| Unconnected Pads | 0 |

## Zone-Connected Nets

- +3V3
- GND

# Cost Estimate

| Metric | Per Board (estimated) |
|---|---|
| PCB Fabrication | ~3.26 USD |
| Components (estimated) | ~4.43 USD |
| Assembly (estimated) | ~2.17 USD |
| **Total (estimated)** | **~9.86 USD** |
| Batch Quantity | 5 |
| Batch Total (estimated) | ~49.28 USD |